

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



UNIVERSITÀ
DI TORINO

Tesi di Laurea Magistrale

The ideally exact categories

A non-pointed generalization
of the notion of semiabelian category

Relatore

professor Alan Cigoli

Candidato

Christian D'Ascoli

Anno Accademico 2024-2025

Table of Contents

Introduction	1
1 Preliminaries	2
2 Theory	5
2.1 Connected limits	5
2.2 The pullback functor and its adjunction	7
2.3 The monad associated to the adjunction	14
Appendix	17

Introduction

Chapter 1

Preliminaries

In order to understand the notions introduced in this thesis is important to recall some definitions and constructions. We also take this opportunity to introduce and fix some notation.

First we define some categories that will be used throughout the thesis. Let $\mathcal{A}$ be a category, we defines the following categories.

- The slice category: given an object B in a category $\mathcal{A}$, we call $(\mathcal{A} \downarrow B)$ the category of objects over E , we denote an object by $(A, \alpha : A \rightarrow B)$. A morphism between two objects (A, α) and (A', α') is a morphism $f : A \rightarrow A'$ such that $\alpha' f = \alpha$.

$$\begin{array}{ccc} A & & A \xrightarrow{f} A' \\ \downarrow \alpha & & \downarrow \alpha \swarrow \alpha' \\ B & & B \end{array}$$

- The coslice category: given an object B in a category $\mathcal{A}$, we call $(B \downarrow \mathcal{A})$ the category of objects under B , we denote an object by $(\alpha : B \rightarrow A, A)$. A morphism between two objects (α, A) and (α', A') is a morphism $f : A \rightarrow A'$ such that $f\alpha = \alpha'$.

$$\begin{array}{ccc} B & & B \\ \downarrow \alpha & & \downarrow \alpha \searrow \alpha' \\ A & & A \xrightarrow{f} A' \end{array}$$

- The category of points: given an object B in a category $\mathcal{A}$, we call $\text{Pt}(B)$ the category of points of B , we denote an object by (A, B, p, s) which is a diagram of the type $A \begin{smallmatrix} \xrightarrow{p} \\ \xleftarrow{s} \end{smallmatrix} B$ with the condition $ps = 1_B$. Morphisms between

$$\begin{array}{ccc}
 A & \xrightarrow{f} & A' \\
 \swarrow p & & \swarrow p' \\
 & B & \\
 \searrow s & & \searrow s'
 \end{array}$$

$$f^* : \text{Pt}(B) \rightarrow \text{Pt}(A)$$

The diagram illustrates the relationship between various maps and objects in a categorical setting. It includes objects A , B , B' , $B' \times_B A$, and A . Maps include s , p , f , f^* , proj_1 , proj_2 , and a curved map sf . A dashed arrow represents the map $\langle 1_{B'}, sf \rangle$.

The diagram illustrates the relationship between a pullback square and its image under a functor f^* . The left square is a pullback with vertices A , A' , B , and B' (implied). The maps are $\alpha: A \rightarrow A'$, $p: A \rightarrow B$, $p': A' \rightarrow B$, $s: B \rightarrow A$, and $s': B \rightarrow A'$. The right square is the image of the left square under f^* , with vertices $B' \times_B A$, A , A' , B' , and B . The maps are $\text{proj}_2: B' \times_B A \rightarrow A$, $\alpha: A \rightarrow A'$, $p: A \rightarrow B$, $p': A' \rightarrow B$, $s: B \rightarrow A$, $s': B \rightarrow A'$, $\text{proj}'_2: B' \times_B A' \rightarrow A'$, $\text{proj}'_1: B' \times_B A' \rightarrow B'$, and $f: B \rightarrow B'$. A dashed arrow $f^*(\alpha)$ connects the top vertices $B' \times_B A$ and $B' \times_B A'$. Curved arrows labeled $\langle 1_{B'}, sf \rangle$ and $\langle 1_{B'}, s'f \rangle$ connect the bottom vertices $B' \times_B A$ and $B' \times_B A'$ to B' .

Definition 1.0.3. *A finitely complete category $\mathcal{A}$ is said to be protomodular if for every morphism $f : A \rightarrow B$ in $\mathcal{A}$, the change-of-base functor $f^* : \text{Pt}(B) \rightarrow \text{Pt}(A)$ is conservative.*

We conclude recalling the following.

Definition 1.0.4. *A category $\mathcal{A}$ is said to be semiabelian if*

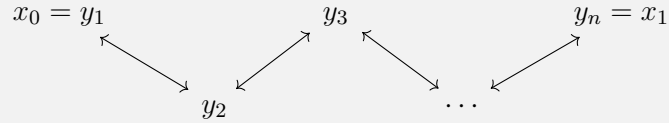
- *It is pointed;*
- *It is exact;*
- *It is protomodular;*
- *It has binary coproducts.*

Chapter 2

Theory

2.1 Connected limits

Definition 2.1.1. A connected limit is a limit over a connected category. A category is connected if it is inhabited and any two objects can be connected by a zigzag of morphisms



where each symbol $\leftrightarrow$ is a morphism $y_i \rightarrow y_{i+1}$ or $y_{i+1} \rightarrow y_i$.

Example 2.1.2. Some examples

- The pullback is a connected limit.

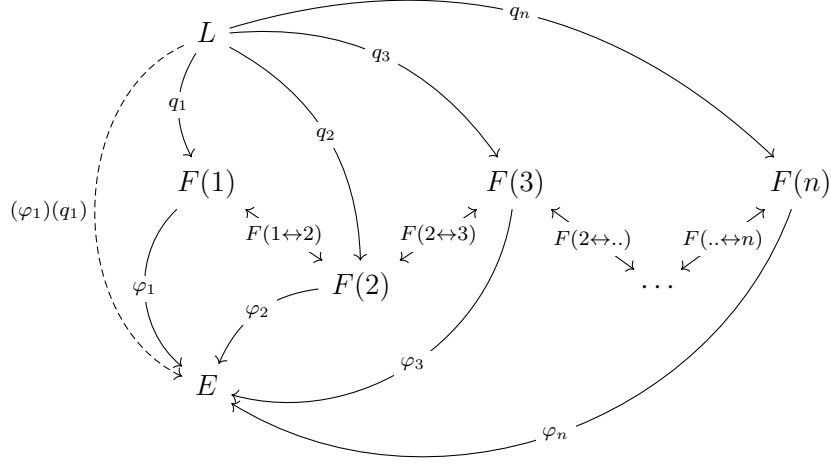
$$\begin{array}{ccc}
 & 2 & \\
 & \downarrow j & \\
 1 & \xrightarrow{i} & 3
 \end{array}$$

- The equalizer is a connected limit.

$$\begin{array}{ccc}
 1 & \xrightarrow{i} & 2 \\
 & \xrightarrow{j} &
 \end{array}$$

Remark 2.1.3. The connected limits in the category $(\mathcal{A} \downarrow E)$ can be computed in $\mathcal{A}$. In fact we can provide to the object L of the limit cone $(L, (q_i)_{i \in \mathcal{I}})$ in $\mathcal{A}$ a morphism to form an object in $(\mathcal{A} \downarrow E)$. In the zigzag below we chose for example $(\varphi_1)(q_1)$, but we could use whatever morphism of the type $(\varphi_i)(q_i)$. The connection guarantees that all the morphisms $q_1 \dots q_n$ involved in the limit in $\mathcal{A}$ are also morphisms in $(\mathcal{A} \downarrow E)$,

ie $(\varphi_i)(q_i) = (\varphi_1)(q_1) \forall i$.



By the previous result, the following holds.

Proposition 2.1.4. *Let E be an object in $\mathcal{A}$. Then the forgetful functor $U : (\mathcal{A} \downarrow E) \rightarrow \mathcal{A}$ creates all connected limits that $\mathcal{A}$ has.*

Note that this result is not true for all the limits. This is extensively shown in the following example.

Example 2.1.5. *The product is not a connected limit, in fact as we can observe the following category is not connected.*

1 2

We try to compute the product directly in $(\mathcal{A} \downarrow E)$. Let (D, δ) and (D', δ') be two objects in $(\mathcal{A} \downarrow E)$, the product is given by the following pullback

$$\begin{array}{ccccc}
 & & A \times_B A' & \longrightarrow & A' \\
 & \swarrow & \searrow \alpha' & & \swarrow \\
 A & \xrightarrow{\alpha} & E & & A' \\
 & \searrow \alpha & \downarrow & & \searrow \alpha' \\
 & & E & &
 \end{array}$$

which doesn't correspond to the product $D \times D'$ in $\mathcal{A}$.

2.2 The pullback functor and its adjunction

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow B$ in $\mathcal{A}$, one can consider functors pullback and shrink

$$\begin{array}{ccc}
 & \xrightarrow{p^*} & \\
 (\mathcal{A} \downarrow B) & & (\mathcal{A} \downarrow E) \\
 & \xleftarrow{p_!} &
 \end{array}$$

defined as follows:

- $p^* : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the pullback functor

$$\begin{array}{ccc}
 \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \swarrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \swarrow \alpha' & & \\ & B & & & \\ \text{proj}_1 \swarrow & & \searrow \text{proj}_1' & & \\ E & & E & & \end{array} \\ \text{dashed arrow } p^*(f) = \tilde{f} \text{ from } E \times_B A \text{ to } E \times_B A' \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the shrink functor

$$\begin{array}{ccc}
 \begin{array}{c} D \\ \downarrow \delta \\ E \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \downarrow \delta & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \downarrow \delta & \swarrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{d} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

Proposition 2.2.1. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then $p_!$ preserves connected limits.*

Proof. By Remark 2.1.3 connected limits are computed in $\mathcal{A}$ and $p_!$ doesn't affect objects and morphism in $\mathcal{A}$. $\square$

Proposition 2.2.2. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then*

$$\begin{array}{ccc} & p^* & \\ (\mathcal{A} \downarrow B) & \xrightarrow{\quad} & (\mathcal{A} \downarrow E) \\ & p_! & \end{array} \quad \text{with } p_! \dashv p^*$$

is an adjunction. In particular the unit is given by

$$\begin{aligned} \eta : \text{Id}_{(\mathcal{A} \downarrow E)} &\rightarrow p^* p_! \\ \eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) &\rightarrow (E \times_B D, \text{proj}_1) \end{aligned}$$

$$\begin{array}{ccccc} D & & \xrightarrow{1_D} & & D \\ & \searrow \langle \delta, 1_D \rangle & & & \downarrow p\delta \\ & E \times_B D & \xrightarrow{\quad} & & D \\ & \downarrow \delta & & & \downarrow p \\ & E & \xrightarrow{p} & & B \end{array}$$

and the counit is given by

$$\begin{aligned} \epsilon : p_! p^* &\rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\ \epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) &\rightarrow (A, \alpha) \end{aligned}$$

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B. \end{array}$$

Proof. Let us show that this is actually an adjunction. First we want to prove that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc} D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\ \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' \end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

$$\begin{array}{ccc} D \xrightarrow{g} D' & & \\ \downarrow \delta \quad \swarrow \delta' & \xrightarrow{p_!} & D \xrightarrow{g} D' \\ E & & \searrow p\delta \quad \downarrow p\delta' \\ & & B \end{array}$$

$$\begin{array}{c}
 \xrightarrow{p^*} \quad \begin{array}{c}
 \begin{array}{ccccc}
 & & \tilde{g} & & \\
 & \swarrow & \text{dashed} & \searrow & \\
 E \times_B D & \xrightarrow{\text{proj}_2} & D & \xrightarrow{g} & D' & \xleftarrow{\text{proj}'_2} & E \times_B D' \\
 \nwarrow \eta_{(D,\delta)} & & \searrow p\delta & & \swarrow p\delta' & & \nwarrow \eta_{(D',\delta')} \\
 & & B & & & & \\
 \nwarrow \text{proj}_1 & & \uparrow p & & \swarrow \text{proj}'_1 & & \\
 & & E & & & &
 \end{array}
 \end{array}
 \end{array}$$

Considering the diagram

$$\begin{array}{ccccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
 \downarrow g & \searrow \text{dashed} & \downarrow p^*p_!(g)=\tilde{g} & & \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
 & & E & \xrightarrow{p} & B
 \end{array}$$

we can prove the naturality in the left diagram by the universal property of the pullback in the right. Since $(\text{proj}_1, \text{proj}_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(f) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. So the naturality condition of η is satisfied.

Now we want to prove that the counit is a natural transformation. Consider $f : (A, \alpha) \rightarrow (A', \alpha')$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & A \\
 p_!p^*(f)=\tilde{f} \downarrow & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A'
 \end{array}$$

As similarly done above, we need to expand the action of the functor $p_!p^*$ on the morphism f , we have:

$$\begin{array}{ccc}
 \begin{array}{c} A \xrightarrow{f} A' \\ \downarrow \alpha \quad \swarrow \alpha' \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & \searrow \text{proj}_1' & \\ & & B & & & & \\ & \swarrow \text{proj}_1 & & \nwarrow p & & \swarrow & \\ & & E & & & & \end{array} \\ \text{proj}_1 \quad \quad \quad \text{proj}_1' \end{array} \\
 \xrightarrow{p!} & & \begin{array}{c} \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & \searrow p(\text{proj}_1') & \\ & & B & & & & \\ & \swarrow p(\text{proj}_1) & & \nwarrow & & \swarrow & \\ & & E & & & & \end{array} \\ p(\text{proj}_1) \quad \quad \quad p(\text{proj}_1') \end{array}
 \end{array}$$

Since $\epsilon_{(A,\alpha)}$ is the $\text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. So also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$, we have:

$$((p^* \epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^* \epsilon)_{(A,\alpha)} (\eta p^*)_{(A,\alpha)} = p^* (\text{proj}_{2,E \times_B A}) (\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\text{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_{1,E \times_B A}} & A \\ & \searrow \alpha(\text{proj}_{2,E \times_B A}) & \swarrow \alpha \\ & & B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} E \times_B (E \times_B A) & \xrightarrow{\text{proj}_{2,E \times_B (E \times_B A)}} & E \times_B A & \xrightarrow{\text{proj}_{2,E \times_B A}} & A & \xleftarrow{\text{proj}_{2,E \times_B A}} & E \times_B A \\ & \searrow (\alpha)(\text{proj}_{2,E \times_B A}) & & \swarrow \alpha & & \searrow \text{proj}_{1,E \times_B A} & \\ & & B & & & & \\ & \swarrow \text{proj}_{1,E \times_B (E \times_B A)} & & \nwarrow p & & \swarrow & \\ & & E & & & & \end{array} \\ \text{proj}_{2,E \times_B A} \end{array} \\
 \circledast = (\tilde{\text{proj}}_2)(\eta_{(E \times_B A, \text{proj}_1)})
 \end{array}$$

and then considering the definition of η , we obtain

$$\begin{array}{ccccc}
 & & & 1_{E \times_B A} & \\
 & & & \curvearrowright & \\
 E \times_B A & & & & E \times_B A \\
 & \searrow \langle \text{proj}_1, E \times_B A, 1_{E \times_B A} \rangle & & & \downarrow p(\text{proj}_1, E \times_B A) \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \\
 & \downarrow & & \downarrow & \\
 \text{proj}_1, E \times_B A & \searrow & E & \xrightarrow{p} & B
 \end{array}$$

$$\circledast = (\text{proj}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle)$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.

The diagram illustrates the following commutative relationships:

- Top Row:** $E \times_B (E \times_B A) \longrightarrow E \times_B A \xrightarrow{\text{proj}_{2, E \times_B A}} A \longleftarrow E \times_B A$
- Bottom Row:** $E \times_B A \longrightarrow B \xrightarrow{p} E$
- Left Curved Arrow:** $E \times_B A \xrightarrow{\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle} E \times_B (E \times_B A)$
- Right Curved Arrow:** $E \times_B A \xrightarrow{\text{proj}_{1, E \times_B A}} E$
- Other Maps:**
 - $1_{E \times_B A} : E \times_B A \rightarrow E \times_B (E \times_B A)$ (top left)
 - $\text{proj}_{2, E \times_B A} : E \times_B A \rightarrow A$ (top middle)
 - $\alpha : A \rightarrow B$ (middle right)
 - $(\alpha)(\text{proj}_{2, E \times_B A}) : E \times_B A \rightarrow B$ (middle left)
 - $\text{proj}_{1, E \times_B (E \times_B A)} : E \times_B (E \times_B A) \rightarrow E$ (bottom left)

$$\circledast = (\mathrm{proj}_2)(\langle \mathrm{proj}_1, E \times_B A, 1_{E \times_B A} \rangle) = \mathrm{Id}_{(E \times_B A, \mathrm{proj}_1)}$$

We want to check that composition of the projections of $E \times_B A$ with $\otimes$ is the same of composing those projection with $\text{Id}_{(E \times_B A, \text{proj}_1)}$. Following the first composition

$$\begin{aligned} & (\mathrm{proj}_{2,E \times_B A})(\mathrm{proj}_2)(\langle \mathrm{proj}_1, 1_{E \times_B A} \rangle) = \\ & = (\mathrm{proj}_{2,E \times_B A})(\mathrm{proj}_{2,E \times_B (E \times_B A)})(\langle \mathrm{proj}_1, 1_{E \times_B A} \rangle) = \\ & = (\mathrm{proj}_{2,E \times_B A})(1_{E \times_B A}) = \mathrm{proj}_{2,E \times_B A} = (\mathrm{proj}_{2,E \times_B A})(\mathrm{Id}_{(E \times_B A, \mathrm{proj}_1)}) \end{aligned}$$

and the second composition

$$(\mathrm{proj}_{1,E \times_B A})(\tilde{\mathrm{pr}}_2)(\langle \mathrm{proj}_1, 1_{E \times_B A} \rangle) = (\mathrm{proj}_{1,E \times_B (E \times_B A)})(\langle \mathrm{proj}_1, 1_{E \times_B A} \rangle) =$$

$$\mathrm{proj}_{1,E \times_B A} = (\mathrm{proj}_{1,E \times_B A})(\mathrm{Id}_{(E \times_B A, \mathrm{proj}_1)})$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned} ((\epsilon p!) \circ (p! \eta))_{(D, \delta)} &= ((\epsilon p!)_{(D, \delta)} (p! \eta))_{(D, \delta)} = (\epsilon_{p!(D, \delta)}) p! (\langle \delta, 1_D \rangle) = \\ &= (\epsilon_{(D, p\delta)}) p! (\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D} (\langle \delta, 1_D \rangle) = 1_D = \text{Id}_{(D, \delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

So we can conclude that the adjunction holds. $\square$

Remark 2.2.3. p^* is a right adjoint functor, so it preserves limits. $p!$ is a left adjoint functor, so it preserves colimits.

Property 2.2.4. The diagram constructed by p^* acting on a morphism $f : (A, \alpha) \rightarrow (A', \alpha')$ is a pullback in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 \\ A & \xrightarrow{f} & A' \end{array}$$

Proof. We prove it in $\mathcal{A}$, then by Remark 2.1.3 it holds in $\mathcal{A} \downarrow B$. Since the right and the total diagrams below are pullback, then the left diagram is a pullback.

$$\begin{array}{ccccc} & & \text{proj}_1 & & \\ & \nearrow & & \searrow & \\ E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' & \xrightarrow{\text{proj}'_1} & E \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 & & \downarrow p \\ A & \xrightarrow{f} & A' & \xrightarrow{\alpha'} & B \\ & \searrow & & \nearrow & \\ & & \alpha & & \end{array}$$

$\square$

We need the following definition in order to claim some properties.

Definition 2.2.5. Let γ be a natural transformation between two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, we say that γ is a cartesian natural transformation if for every morphism

$f : A \rightarrow B$ in $\mathcal{C}$, the following diagram is a pullback:

$$\begin{array}{ccc} F(A) & \xrightarrow{\gamma_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\gamma_B} & G(B) \end{array}$$

ie the naturality diagrams are pullback.

Property 2.2.6. The unit η and the counit ϵ are cartesian.

Proof. First we prove that η is cartesian. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$. Applying *Property 2.2.4* to $p_!(g)$, we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\ E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \end{array}$$

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left diagram is a pullback, as stated.

$$\begin{array}{ccccc} & & 1_D & & \\ & \nearrow & & \searrow & \\ D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\ & \nwarrow & & \nearrow & \\ & & 1_{D'} & & \end{array}$$

Now we want to prove that ϵ is cartesian. Let $f : (A, \alpha) \rightarrow (A', \alpha')$ be a morphism in $(\mathcal{A} \downarrow B)$. By *Proposition 2.2.1* applied at the pullback in *Property 2.2.4* for f we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\epsilon_{(A, \alpha)} = \text{proj}_2} & A \\ \downarrow p_! p^*(f) = \tilde{f} & & \downarrow f \\ E \times_B A' & \xrightarrow{\epsilon_{(A', \alpha')} = \text{proj}'_2} & A' \end{array}$$

□

Proposition 2.2.7. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then p^* preserves coequalizer of kernel pairs whose coequalizers are pullback stable regular epimorphisms.*

Proof. Consider the following diagram

$$\begin{array}{ccccc}
 C & & & & C' \\
 \exists! s \downarrow & \nearrow r_2 & & \nearrow u & \exists! t \downarrow \\
 p^*(\text{Eq}(q)) & \xrightarrow[\tilde{p}_1]{\tilde{p}_2} & p^*(A) = A \times_B E & \xrightarrow{\tilde{q}} & p^*(\text{Coeq}(p_1, p_2)) \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Eq}(q) & \xrightarrow[p_1]{p_2} & A & \xrightarrow{q} & \text{Coeq}(p_1, p_2)
 \end{array}$$

Let q be the coequalizer of the kernel pair $\text{Eq}(q)$. We can affirm by *Property 2.2.4* applied to q that the diagram in the bottom-right corner is a pullback. Since by hypothesis q is a pullback stable regular epimorphism, we have that $\tilde{q}$ is a regular epimorphism, ie it is the coequalizer of some parallel arrows. Let (r_1, r_2) be such pair.

Now we want to prove that $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. Since $\tilde{q}$, $\tilde{p}_1$ and $\tilde{p}_2$ are obtained applying p^* to the kernel pair $\text{Eq}(q)$, and p^* preserves limits, then $p^*(\text{Eq}(q))$ is the kernel pair of $\tilde{q}$. So by the universal property there exists a unique morphism $s : C \rightarrow p^*(\text{Eq}(q))$ such that $\tilde{p}_1 s = r_1$ and $\tilde{p}_2 s = r_2$. Now let u be a morphism such that $u\tilde{p}_1 = u\tilde{p}_2$. We have that $ur_1 = u\tilde{p}_1 s = u\tilde{p}_2 s = ur_2$, so by the universal property of the coequalizer $\tilde{q}$ there exists a unique morphism $t : p^*(\text{Coeq}(p_1, p_2)) \rightarrow C'$ such that $t\tilde{q} = u$. So $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. $\square$

2.3 The monad associated to the adjunction

Consider the monad associated to the adjunction we just studied, we want to study some properties that will be useful.

Property 2.3.1. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. The monad $(p^*p_!, \eta, p^*\epsilon p_!)$ is cartesian.*

Proof. We need to prove that the category $(\mathcal{A} \downarrow E)$ has pullback, the monad preserves pullbacks, and both the multiplication $p^*\epsilon p_!$ and the unit η are cartesian.

The category $(\mathcal{A} \downarrow E)$ has pullbacks, by *Proposition 2.1.4*, since $\mathcal{A}$ has. The monad $p^*p_!$ preserves pullbacks, in fact p^* is a right adjoint so it preserves limits, therefore preserves pullbacks, and $p_!$ preserves pullbacks by *Proposition 2.2.1*. We already proved that the unit η is cartesian, so it remains to prove that the multiplication $p^*\epsilon p_!$ is cartesian. Since ϵ is cartesian and p^* preserves pullbacks as shown above, we can conclude that the multiplication is cartesian. $\square$

Definition 2.3.2. A monad (T, η, μ) on a category $\mathcal{X}$ with finite coproducts is said to be essentially nullary if the morphism $[T(!_X), \eta_X]$ is a strong epimorphism for each object X in $\mathcal{X}$, where $!_X$ is the unique morphism from the initial object of $\mathcal{X}$ to X .

Now we claim a lemma which will be useful to prove some theorems.

Lemma 2.3.3. If a category $\mathcal{X}$ is protomodular, then for each commutative diagram of the form

$$\begin{array}{ccc} B' \times_B A & \xrightarrow{u} & A \\ q \downarrow \uparrow t & & p \downarrow \uparrow s \\ B' & \xrightarrow{v} & B \end{array}$$

where p, q are split epimorphisms with respective sections s, t and the downward directed square is a pullback, it holds that the pair (u, s) is a strongly epimorphic pair.

Proof. Assume $\mathcal{X}$ is protomodular. We want to prove that if u and s factorize through a mono m as $u = mf$ and $s = ms'$, then m is an iso. By (cited appendice), this implies that (u, s) is a strongly epimorphic pair. Define $p' = pm$, we have that $p's' = 1_Y$, in fact $p's' = pms' = ps = 1_Y$. Pulling back with the change-of-base functor v^* the mono m along v we obtain the following diagram. In this operation we also generate the morphism q', t', w . In the same way as in Property 2.2.4 we have that the diagram between m and $v^*(m)$ is a pullback.

□

Theorem 2.3.4. If $\mathcal{X}$ is a pointed protomodular category with finite coproducts and (T, η, μ) is a monad on $\mathcal{X}$ with cartesian η , then the monad is essentially nullary.

Proof. For each object X in $\mathcal{X}$ we can consider the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & T(X) \\ !^X \downarrow \uparrow !_X & & T(!_X) \downarrow \uparrow T(!_X) \\ 0 & \xrightarrow{\eta_0} & T(0) \end{array}$$

Since $\mathcal{X}$ is protomodular then by the lemma above, the pair $(\eta_X, T(!_X))$ is a strongly epimorphic. It is a standard fact that given f and g morphism, if (f, g) is a strongly epimorphic pair, then $[f, g]$ is a strong epi. So we conclude that $[\eta_X, T(!_X)]$ is a strong epi, ie the monad is essentially nullary. (citazione) $\square$

Appendix

Definition A.0.5. FSF_kDTU_{InT}